

Data Analysis with Python

Cheat Sheet: Exploratory Data Analysis

Package/Method	Description	Code Example
Complete dataframe correlation	Correlation matrix created using all the attributes of the dataset.	<pre>df.corr()</pre>
Specific Attribute correlation	Correlation matrix created using specific attributes of the dataset.	<pre>df[['attribute1','attribute2',...]].corr()</pre>
Scatter Plot	Create a scatter plot using the data points of the dependent variable along the x-axis and the independent variable along the y-axis.	<pre>from matplotlib import pyplot as plt plt.scatter(df[['attribute_1']],df[['attribute_2']])</pre>
Regression Plot	Uses the dependent and independent variables in a Pandas data frame to create a scatter plot with a generated linear regression line for the data.	<pre>import seaborn as sns sns.regplot(x='attribute_1',y='attribute_2', data=df)</pre>
Box plot	Create a box-and-whisker plot that uses the pandas dataframe, the dependent, and the independent variables.	<pre>import seaborn as sns sns.boxplot(x='attribute_1',y='attribute_2', data=df)</pre>
Grouping by attributes	Create a group of different attributes of a dataset to create a subset of the data.	<pre>df_group = df[['attribute_1','attribute_2',...]]</pre>
GroupBy statements	a. Group the data by different categories of an attribute, displaying the average value of numerical attributes with the same category.	<pre>a) df_group = df.groupby(['attribute_1'],as_index=False).mean() b) df_group = df.groupby(['attribute_1','attribute_2'],as_index=False).mean()</pre>

	b. Group the data by different categories of multiple attributes, displaying the average value of numerical attributes with the same category.	
Pivot Tables	Create Pivot tables for better representation of data based on parameters	<code>grouped_pivot = df_group.pivot(index='attribute_1',columns='attribute_2')</code>
Pseudocolor plot	Create a heatmap image using a PsuedoColor plot (or pcolor) using the pivot table as data.	<code>from matplotlib import pyplot as plt plt.pcolor(grouped_pivot, cmap='RdBu')</code>
Pearson Coefficient and p-value	Calculate the Pearson Coefficient and p-value of a pair of attributes	<code>From scipy import stats pearson_coef,p_value=stats.pearsonr(df['attribute_1'],df['attribute_2'])</code>

